Arista Access Node WHI-ACC-61 QoS Congestion Event (ALARM_9917) Resolution SOP

Node: WHI-ACC-61 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_9917

Description

This Standard Operating Procedure (SOP) outlines the diagnostic and resolution procedures for a critical QoS congestion event (microbursts, interface tail drops, QoS queue starvation) on an Arista Access node (Role: Access) at the Whitefield East POP (Node ID: WHI-ACC-61). The underlying cause of this failure domain is rooted in the interplay between the node's buffer sizes, the ingress traffic patterns, and the QoS policy configuration. The Arista node's buffer sizes are insufficient to handle the short-term microburst traffic spikes, leading to interface tail drops and subsequent QoS queue starvation. This congestion event triggers the ALARM_9917 alarm, signaling a critical failure in the network.

Pre-requisite Checks

- Verify the redundant link status is operational and not contributing to the congestion event.

- Ensure the node's buffer sizes are set according to the vendor's recommendations and the node's capacity.

- Review the QoS policy configuration to ensure it is optimized for the node's capacity and traffic patterns.

Potential Root Causes

Cause	Probability
Insufficient buffer sizes to handle microburst traffic spikes	High
Inefficient QoS policy configuration	Medium
Hardware failure or misconfiguration	Low

Diagnostic Commands & Expected Outputs

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Command: show interface detail, Expected Output: Interface statistics, including packet drops and QoS queue utilization., Description: This command provides an overview of the interface's statistics, including packet drops, which can indicate congestion events.

Command: show qos policy, Expected Output: QoS policy configuration, including buffer sizes and queue settings., Description: This command displays the QoS policy configuration, allowing the operator to verify if the buffer sizes and queue settings are optimized for the node's capacity and traffic patterns.

Command: show buffer detail, Expected Output: Buffer sizes and utilization statistics., Description: This command provides detailed
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information about the node's buffer sizes and utilization statistics, helping the operator to identify if the buffer sizes are sufficient to handle the traffic spikes.
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Resolution Steps

Step 1: Step: Increase buffer sizes to handle microburst traffic spikes, Command: buffer size , Description: Increase the buffer size to handle the short-term microburst traffic spikes, ensuring the node can process the traffic without dropping packets.

Step 2: Step: Optimize QoS policy configuration, Command: qos policy buffer-size , Description: Adjust the QoS policy configuration to optimize buffer sizes and queue settings according to the node's capacity and traffic patterns, ensuring efficient traffic processing and minimizing congestion events.

Step 3: Step: Verify and adjust redundant link settings, Command: link , Description: Verify the redundant link settings and adjust as necessary to ensure the link is operational and not contributing to the congestion event.

Post-Check Verification

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Command: show interface detail, Expected Output: No packet drops or QoS queue utilization above 50%., Description: Verify that the interface statistics indicate no packet drops and QoS queue utilization is below 50%, ensuring the congestion event has been resolved.

Command: show qos policy, Expected Output: Optimized QoS policy configuration., Description: Verify that the QoS policy configuration has been optimized for the node's capacity and traffic patterns, ensuring efficient traffic processing.

Command: show buffer detail, Expected Output: Buffer sizes and utilization statistics indicating no congestion events., Description: Verify that the buffer sizes and utilization statistics indicate no congestion events, ensuring the node can process the traffic without dropping packets.
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Roll-back Procedures

- Step: Restore original buffer sizes, Command: buffer size , Description: Restore the original buffer sizes to prevent unintended consequences of the increased buffer size.

- Step: Restore original QoS policy configuration, Command: qos policy buffer-size , Description: Restore the original QoS policy configuration to prevent unintended consequences of the optimized configuration.

- Step: Restore original redundant link settings, Command: link , Description: Restore the original redundant link settings to prevent unintended consequences of the adjusted settings.